

# Intermediate Macroeconomics: Lecture 1 (Ch.1)

ECON 311 – Intermediate Macroeconomics

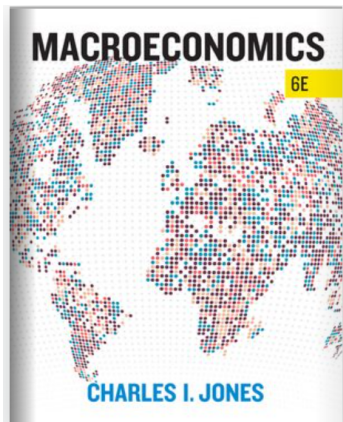
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# Textbook

- We are going to follow the textbook for the majority of the course
- Textbook: *Macroeconomics*, by Charles I. Jones, 5th edition



# What are You Going to Learn (ch. 1)

- What are the important macroeconomics questions to be considered
- The three-part structure of the course/textbook:
  - The long run
  - The short run
  - Issues for the future
- Foundations of macroeconomic models and modeling

# Important Macroeconomic Questions to Consider

Long-run questions:

- Why is today's average North American
  - more than 10 times richer than 100 years ago?
  - 50 times richer than the average Ethiopian?
- What could be the cause of high income inequality across the world and even within a single country?
- Why has the unemployment rate in Europe been almost twice as high as it has been in the United States?

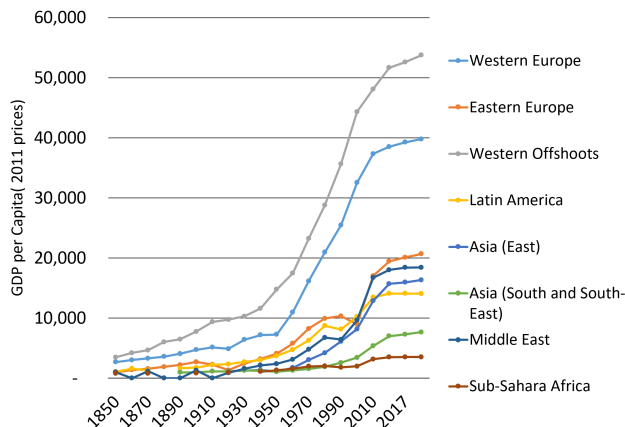
# Important Macroeconomic Questions to Consider

Short-run questions:

- Do we understand and know the causes of the global financial crisis in 2007-2008
  - Was it greedy banks lending money, irresponsible homeowners living beyond their means, or government institutions such as Freddie Mac and Fannie Mae?
- What are the causes of the recent rise in prices?
- How did the 2008 Affordable Care Act impact the price of healthcare, economic growth, and government expenditures?

# GDP Per Capita Across Regions

- GDP Per Capita represents the average income per individual in a country
- GDP per Capita is much higher in developed countries (Western Europe and Western Offshoots - North America & Australia & New Zealand) compared to the rest
- Asia, Eastern Europe and Middle East experience an uptick in recent years

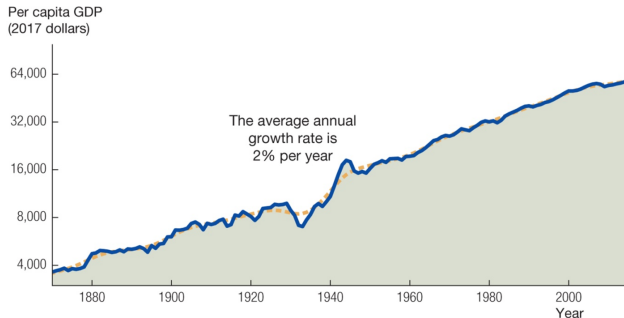


# The Long Run

- Income per person in the United States:
  - \$2,800 in 1870
  - \$44,000 in 2008
  - \$76,000 in 2025
- As seen earlier, many countries have not experienced similar increases in living standards
- The analysis of economic growth helps explain the long run

# Per Capita GDP in the United States, 1870–2015

- Overall growth is upward
- Short-run fluctuations can cause peaks and troughs
- It encourages us to ask, Will there ever be a long-run peak? Can we grow forever? What does the future hold?



Sources: The Maddison-Project, [www.ggd.net/maddison/](http://www.ggd.net/maddison/), and U.S. Department of Commerce, Bureau of Economic Analysis. The blue line shows actual per capita GDP in the United States. The orange line, which is hard to distinguish from the blue line, shows potential output. The difference between actual and potential output is a measure of short-run fluctuations in GDP.

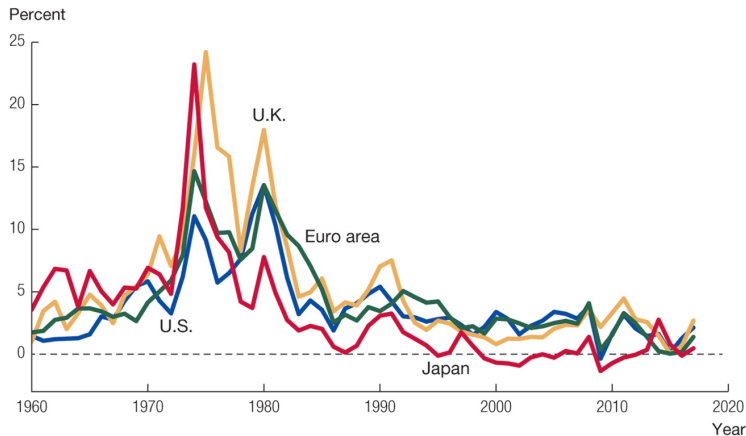
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# The Short Run

- Potential output ( $Y_p$ ): Measure of how per capita GDP would evolve with completely flexible prices and fully employed resources
- Short-run fluctuations are often defined as deviations of actual output ( $Y$ ) from its potential level
  - When  $Y > Y_p$ , this is an expansion
  - When  $Y < Y_p$ , this is a recession
- In 1982, actual output was 5 percent less than potential output, which is economically important
  - In today's prices, the gap was roughly \$1,500 per person (\$6,000 for a typical family of 4)
  - This recession represented a large cost in terms of lost income
- Recessions are costly and cause devastation in the short run but are overcome by long-run trends

# Inflation Rate in Certain Developed Countries

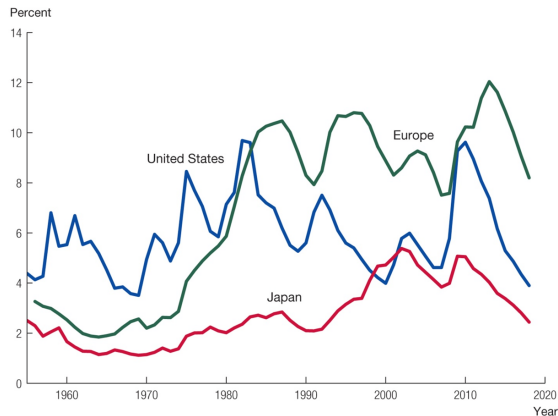
- The inflation rate is the percentage change in the price level – often measured using CPI



Source: FRED. The inflation rate measures the percentage change in consumer prices.  
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# The Unemployment Rate in the United States, Europe, and Japan

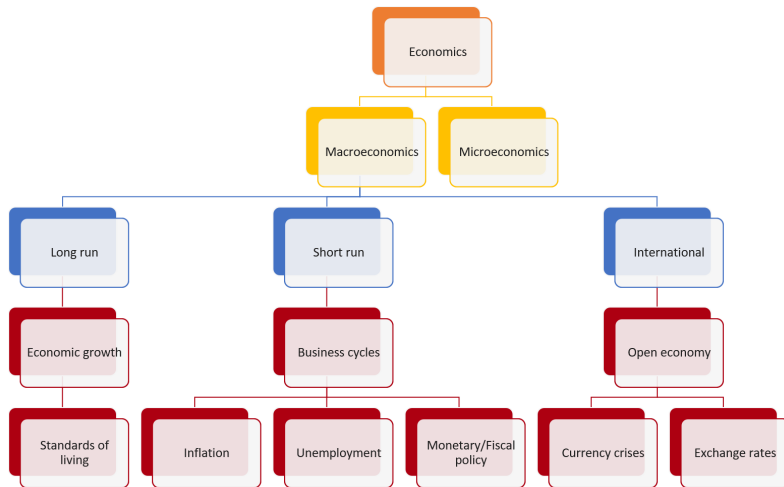
- Unemployment rates are highly correlated with the business cycle
- Before 1980, unemployment rates in Europe were actually lower than in the United States, while Japan has remained low over this time frame



Sources: OECD Main Economic Indicators and the U.S. Bureau of Labor Statistics.  
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# An Overview of the Course

- Microeconomics also has many subfields and a lot of them are overlapped with macroeconomics



# How Macroeconomics Studies Key Questions

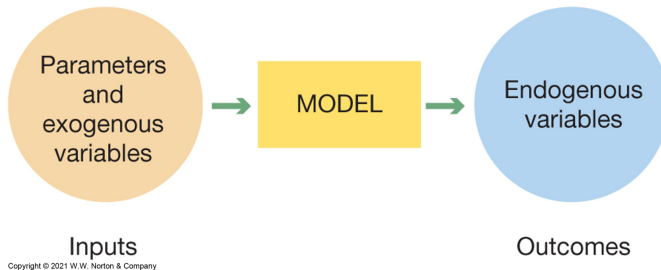
Macroeconomists have a general approach to study questions of interest:

- Document the facts
  - In recent years, more macroeconomists rely more and more on micro-data
- Develop a model
  - Models simplify the complicated real world into its most relevant elements
    - The goal of a model is not to perfectly replicate every single detail/feature of markets or behavior
  - A model is useful if it has good predictive power
  - Economic models often involve systems of multiple equations
- Compare predictions of the model with original facts
- Use the model to make other predictions that will eventually be tested

# Parts of an Economic Model

- Parameter: An input that is fixed over time, except when the model builder changes it for an experiment
  - Minimum we assume not to change; can be changed by the model builder as an experiment
- Exogenous variable: An input that can change over time, but determined ahead of time by the model builder
  - Exogenous = “outside of the model”
  - Can change over time but determined outside of the model.
- Endogenous variable: An outcome of the model—something that is explained by the model
  - Endogenous = “within the model”
  - Determined by factors plugged into our model.

# Parts of an Economic Model



- For example, a linear model

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2$$

# Suppose We Have a Working Model...

- How can we use it?
  - Models are built using existing data, but can be used to predict events and outcomes for which we do not have data yet – set up a counterfactual “what if” situation
  - Change parameters and exogenous variables to see how they affect endogenous variables
  - Predict costs and benefits of new government policies – a change in parameter that mimic the policy
  - All models can make inaccurate predictions, so we should always interpret the results with caution

# Model Example: Labor supply and demand

- Variables:
  - $L_s$  = number of hours laborers want to work
  - $L_D$  = number of labor hours firms want to hire
  - $w$  = wage
- no exogenous variables
- Parameter:  $\bar{f}, \bar{l}, \bar{a}$

# Model Example: Labor supply and demand

- Supply Function:

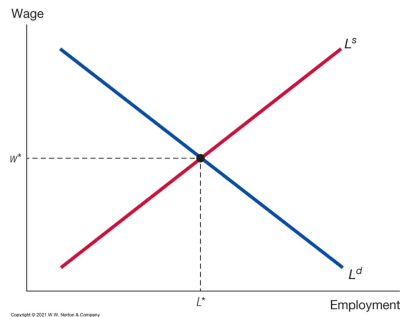
$$L_s = f(w) = \bar{a}w + \bar{l} = 2w + 30$$

- Demand Function:

$$L_d = g(w) = \bar{f} - w = 60 - w$$

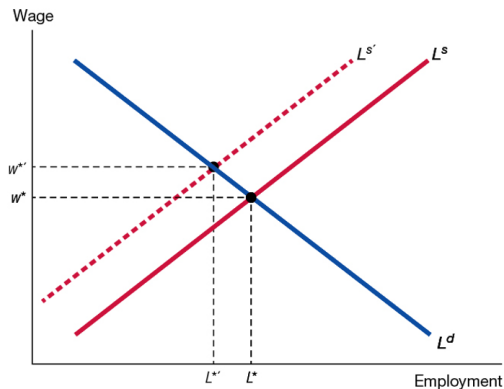
- Equilibrium:

$$L_d = L_s$$



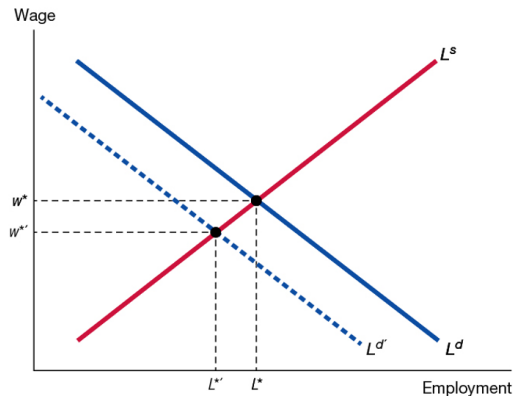
# Model Example: Labor supply and demand

- Consider a case of an increase in income taxes
- Firm would still pay a wage of  $w$ , but worker receives only a wage of  $(1 - t)w$
- For any given wage, the worker gets to keep less of his wage so he supplies less labor. In equilibrium, we move from  $(w^*, L^*)$  to  $(w^{*'}, L^{*'})$



# Model Example: Labor supply and demand

- Consider a case of an increase in an input price
- As production costs increase for firms, their demand for labor decreases. It lead to a leftward shift in  $L_d$
- The equilibrium move from  $(w^*, L^*)$  to  $(w^{*'}, L^{*'})$



# Next Lecture

- Next lecture will cover Ch.2